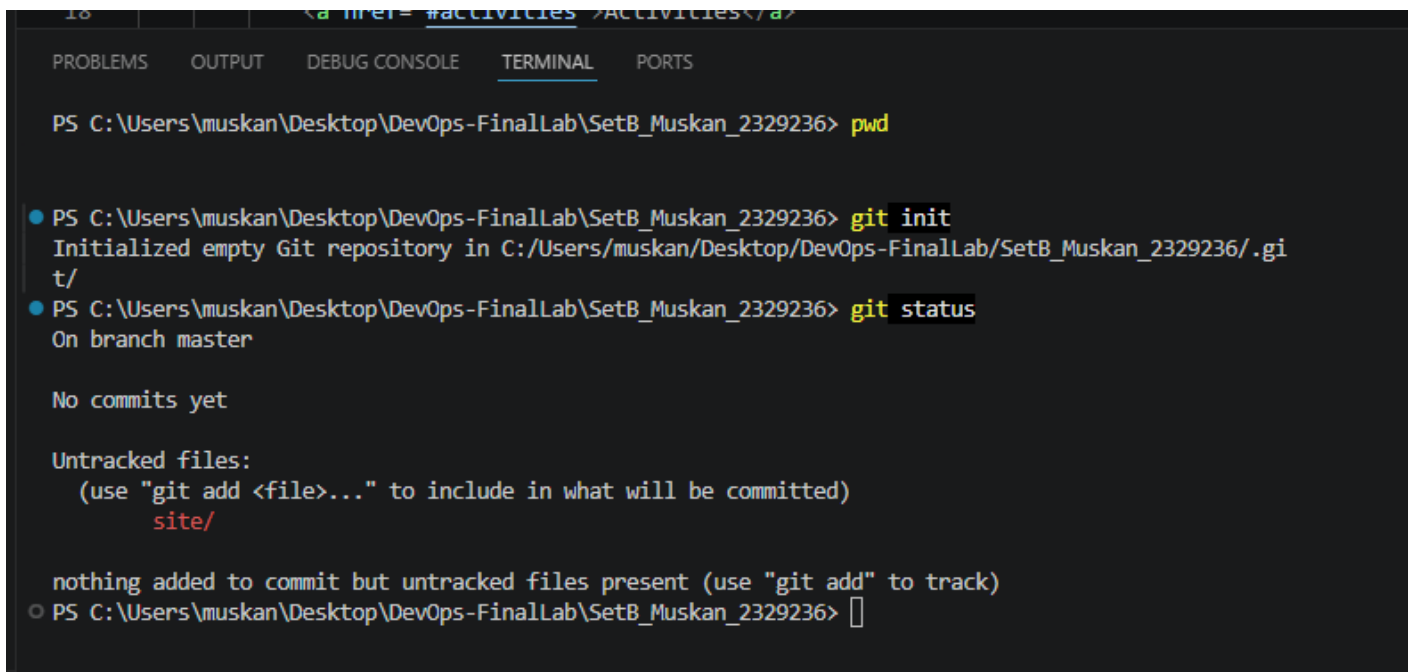
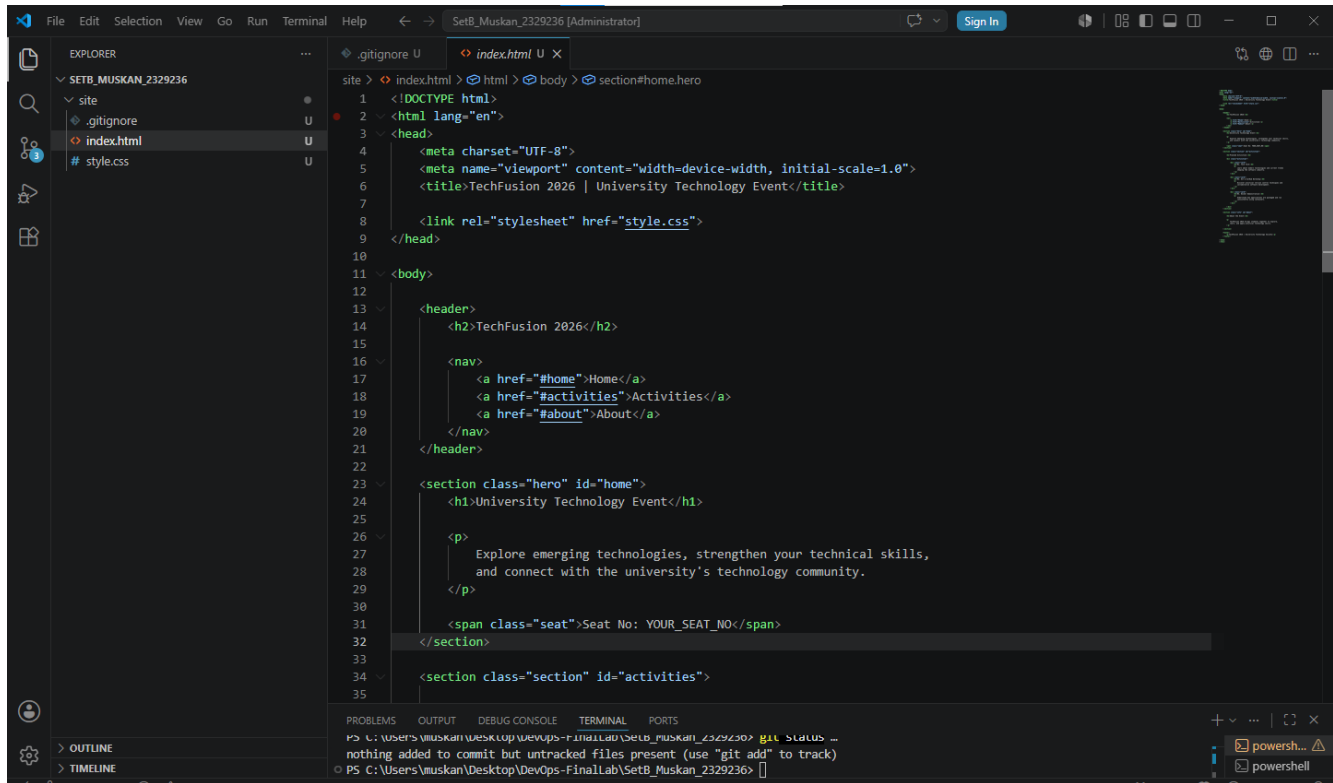


DevOps Final Lab Evidence Screenshots

Muskan Kamran : SE-2329236

Q1(a) — GitHub Repository & Initial Commit



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git status
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git add .
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   site/.gitignore
        new file:   site/index.html
        new file:   site/style.css

○ PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> |
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

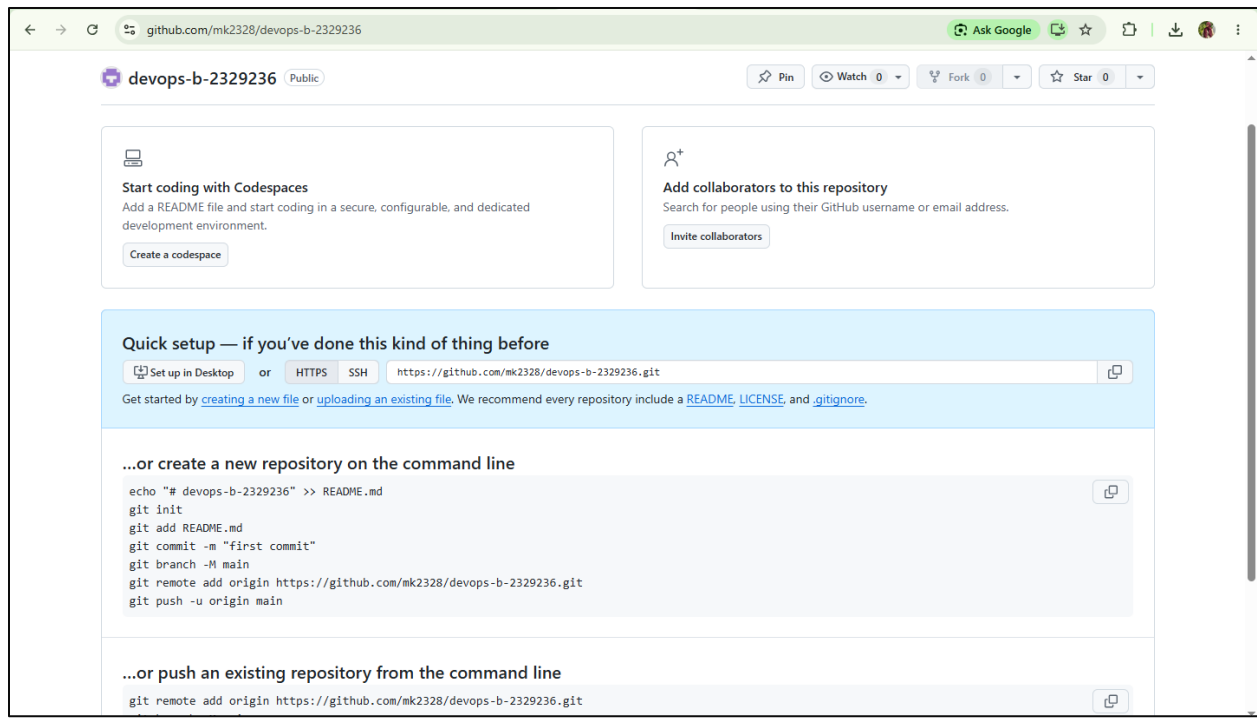
PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git status
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git commit -m "Initial university event website"
[master (root-commit) a94d225] Initial university event website
Committer: exam <exam@aptechnn2.edu>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

3 files changed, 191 insertions(+)
create mode 100644 site/.gitignore
create mode 100644 site/index.html
create mode 100644 site/style.css
○ PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> |
```



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git commit -m "Initial university event website"

After doing this, you may fix the identity used for this commit with:

git commit --amend --reset-author

3 files changed, 191 insertions(+)
create mode 100644 site/.gitignore
create mode 100644 site/index.html
create mode 100644 site/style.css
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git log --oneline
a94d225 (HEAD -> master) Initial university event website
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git branch -M main
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git branch
* main
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git remote add origin https://github.com/mk2328/devops-b-2329236.git
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git remote -v
origin https://github.com/mk2328/devops-b-2329236.git (fetch)
origin https://github.com/mk2328/devops-b-2329236.git (push)
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git push origin main
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 1.68 KiB | 862.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mk2328/devops-b-2329236.git
 * [new branch]      main -> main
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> |
```

Q1(b) — Branch, Commit & Merge

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git push origin main
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 1.68 KiB | 862.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mk2328/devops-b-2329236.git
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git switch -c update-content
Switched to a new branch 'update-content'
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git branch
main
* update-content
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git status
On branch update-content
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   site/index.html

no changes added to commit (use "git add" and/or "git commit -a")

no changes added to commit (use "git add" and/or "git commit -a")
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git commit -m "Add fourth event activity"
[update-content f15bcf3] Add fourth event activity
Committer: exam <exam@aptechnn2.edu>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 10 insertions(+)
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git push origin update-content
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 541 bytes | 135.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'update-content' on GitHub by visiting:
remote:   https://github.com/mk2328/devops-b-2329236/pull/new/update-content
remote:
To https://github.com/mk2328/devops-b-2329236.git
 * [new branch]      update-content -> update-content
💡 PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236>
```

```

* [new branch]      update-content -> update-content
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git switch main
Switched to branch 'main'
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git merge update-content
Updating a94d225..f15bcf3
Fast-forward
 site/index.html | 10 ++++++++
 1 file changed, 10 insertions(+)
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mk2328/devops-b-2329236.git
 a94d225..f15bcf3  main -> main
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git status
On branch main
nothing to commit, working tree clean
○ PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236>

```

Q1(c) — GitHub Actions Deployment

The screenshot shows the Visual Studio Code interface with the Explorer view on the left displaying the file structure of the project. The file `pages.yml` is selected. The main editor area shows the content of `pages.yml`, which is a GitHub Actions workflow for deploying a website to GitHub Pages. The workflow includes steps for checking out the repository, setting up Pages, and uploading the website artifact. The terminal window at the bottom shows the output of the `git status` command, indicating that the working tree is clean and on the `main` branch.

```

site > .github > workflows > ! pages.yml
1  name: Deploy Website to GitHub Pages
2
3  on:
4    push:
5      branches:
6        - main
7
8  permissions:
9    contents: read
10   pages: write
11   id-token: write
12
13  concurrency:
14    group: pages
15    cancel-in-progress: true
16
17  jobs:
18    deploy:
19      runs-on: ubuntu-latest
20
21      environment:
22        name: github-pages
23        url: ${ steps.deployment.outputs.page_url }
24
25      steps:
26        - name: Checkout repository
27          uses: actions/checkout@v4
28
29        - name: Setup Pages
30          uses: actions/configure-pages@v5
31
32        - name: Upload website
33          uses: actions/upload-pages-artifact@v3
34
35  PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
36
37 ● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git status
38 On branch main
39 nothing to commit, working tree clean
40 ○ PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236>

```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

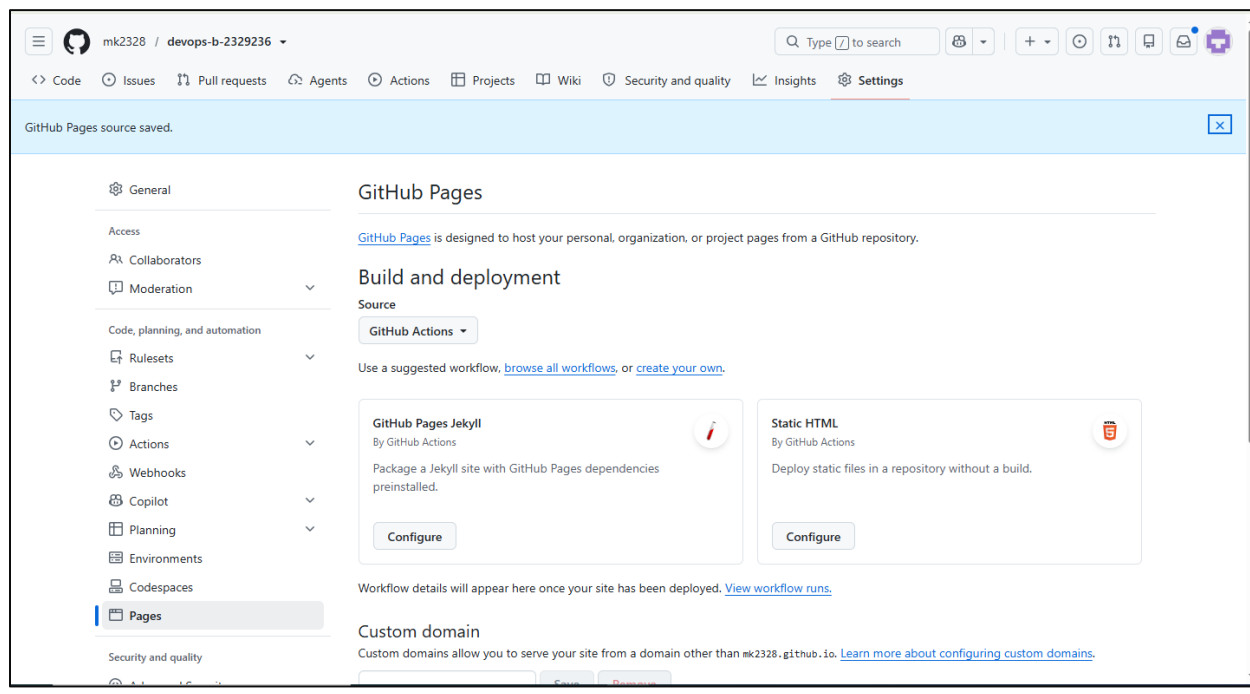
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git add .github/workflows/pages.yml
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git commit -m "Add GitHub Pages deployment workflow"
[main 45050e9] Add GitHub Pages deployment workflow
Committer: exam <exam@aptechnn2.edu>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 39 insertions(+)
create mode 100644 .github/workflows/pages.yml
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git push origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 714 bytes | 89.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mk2328/devops-b-2329236.git
   f15bcf3..45050e9  main -> main
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> |
```



mk2328 / devops-b-2329236

Q Type to search

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pages.yml

2 workflow runs

EventStatusBranchActor

Enable GitHub Pages deployment

Deploy Website to GitHub Pages #2: Commit 011f172 pushed by mk2328

main

1 minute ago18s

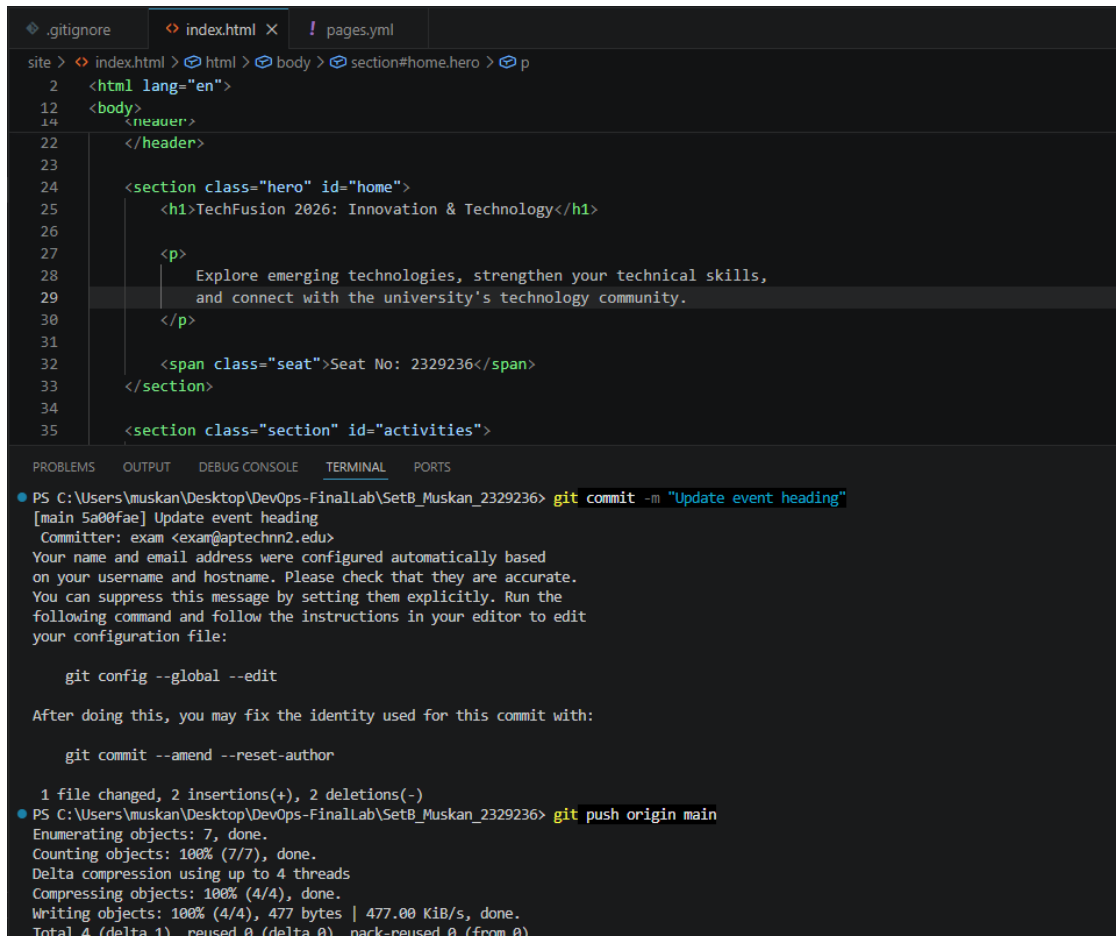
Add GitHub Pages deployment workflow

Deploy Website to GitHub Pages #1: Commit 45050e9 pushed by mk2328

main

7 minutes ago8s

Q1(d) — Updated GitHub Pages Website



The image shows a VS Code editor with a file explorer on the left showing `.gitignore`, `index.html`, and `pages.yml`. The main editor displays the `index.html` file with the following content:

```
2 <html lang="en">
12 <body>
14 <header>
22 </header>
23
24 <section class="hero" id="home">
25 <h1>TechFusion 2026: Innovation & Technology</h1>
26
27 <p>
28   Explore emerging technologies, strengthen your technical skills,
29   and connect with the university's technology community.
30 </p>
31
32 <span class="seat">Seat No: 2329236</span>
33 </section>
34
35 <section class="section" id="activities">
```

The terminal window at the bottom shows the following commands and output:

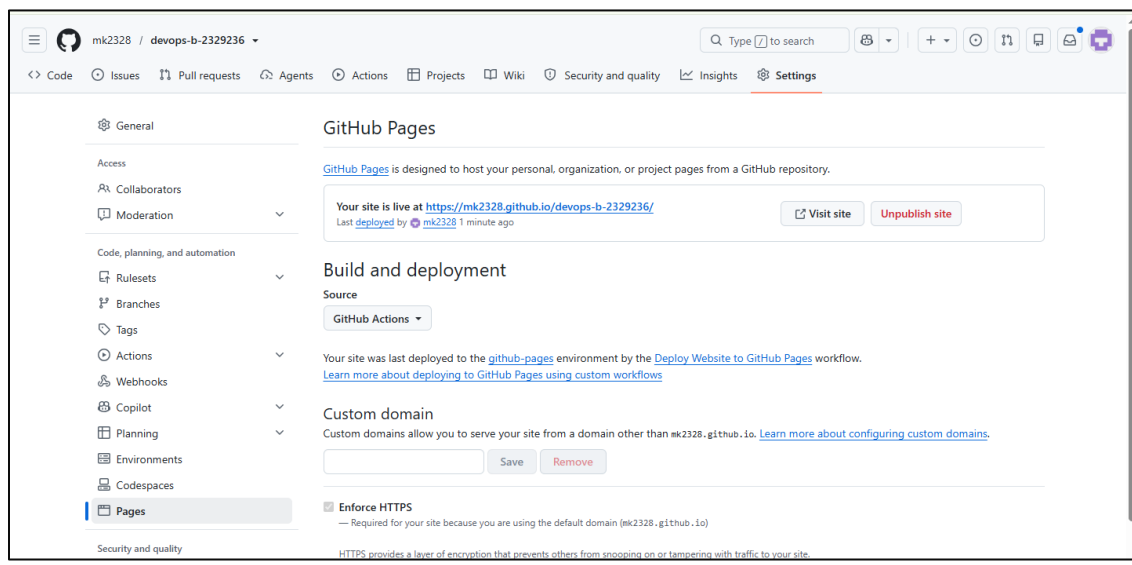
```
PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git commit -m "Update event heading"
[main 5a00fae] Update event heading
Committer: exam <exam@aptechnn2.edu>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

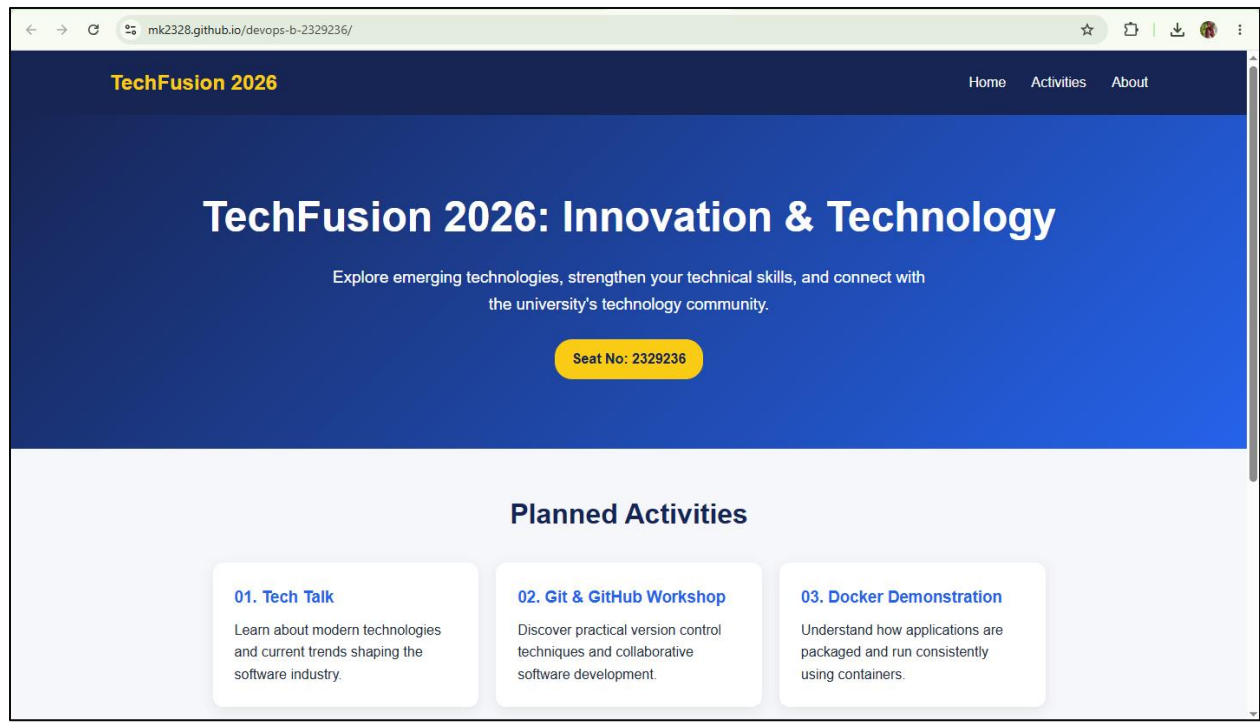
    git commit --amend --reset-author

1 file changed, 2 insertions(+), 2 deletions(-)
PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 477 bytes | 477.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
```

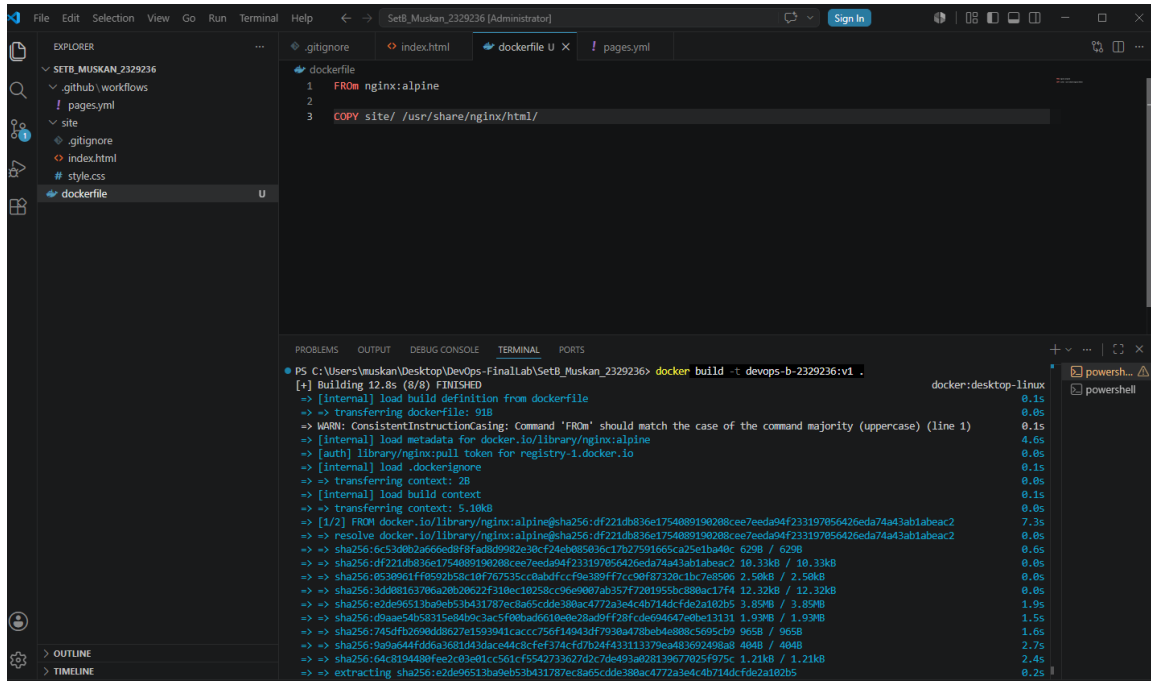


The image shows the GitHub repository settings page for the repository `mk2328 / devops-b-2329236`. The `Pages` tab is selected in the left sidebar. The main content area is titled `GitHub Pages` and contains the following sections:

- General**: The `GitHub Pages` section is expanded, showing that the site is live at <https://mk2328.github.io/devops-b-2329236/>. The last deployment was by `mk2328` 1 minute ago. There are buttons for `Visit site` and `Unpublish site`.
- Build and deployment**: The `Source` is set to `GitHub Actions`. A message states: "Your site was last deployed to the `github-pages` environment by the `Deploy Website to GitHub Pages` workflow. [Learn more about deploying to GitHub Pages using custom workflows](#)".
- Custom domain**: A section for custom domains with a text input field, `Save`, and `Remove` buttons. A message states: "Custom domains allow you to serve your site from a domain other than `mk2328.github.io`. [Learn more about configuring custom domains](#)."
- Enforce HTTPS**: A checkbox is checked, with a message: "Required for your site because you are using the default domain (`mk2328.github.io`). [HTTPS provides a layer of encryption that prevents others from snooping on or tampering with traffic to your site.](#)"



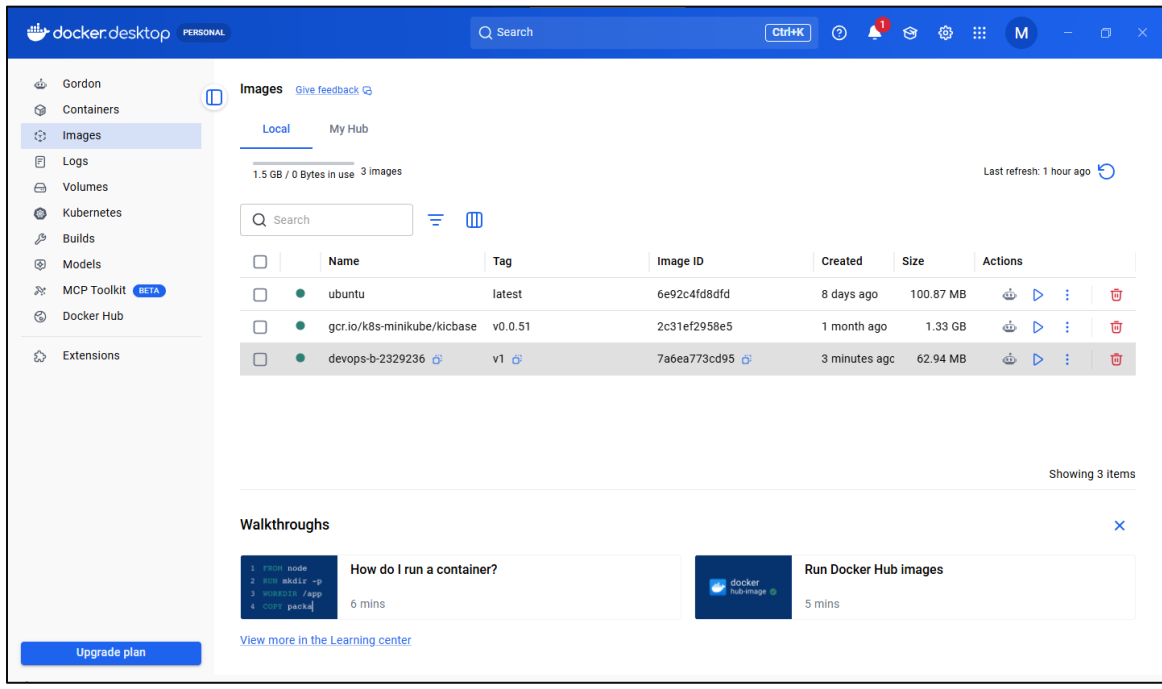
Q2(a) — Docker Image, Container & Local Website



The screenshot shows the Visual Studio Code editor with a file explorer on the left displaying a project named 'SETB_MUSKAN_2329236'. The file explorer shows a directory structure with files like '.gitignore', 'index.html', 'pages.yml', 'site', 'style.css', and 'dockerfile'. The 'dockerfile' file is selected, and its content is displayed in the editor:

```
1 FROM nginx:alpine
2
3 COPY site/ /usr/share/nginx/html/
```

The terminal at the bottom shows the output of the command 'docker build -t devops-b-2329236:v1'. The output indicates that the build is successful and shows the progress of the build process, including the resolution of the 'FROM' instruction and the copying of the 'site' directory to the container.



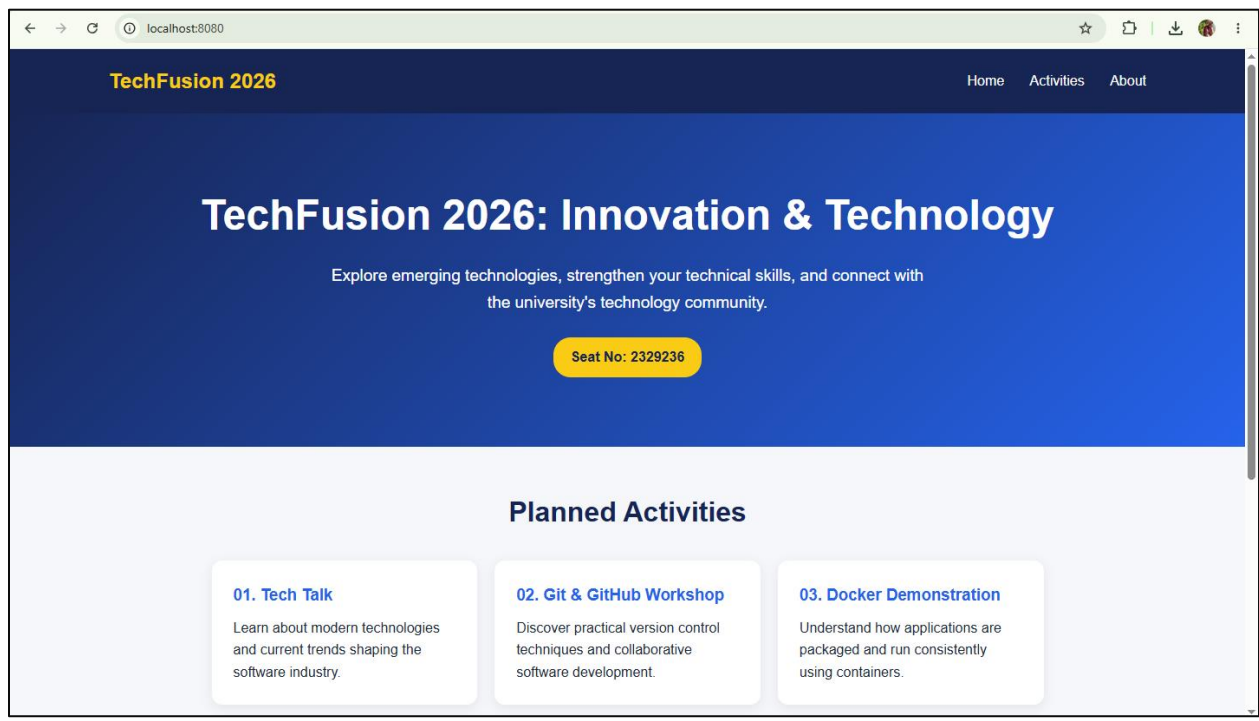
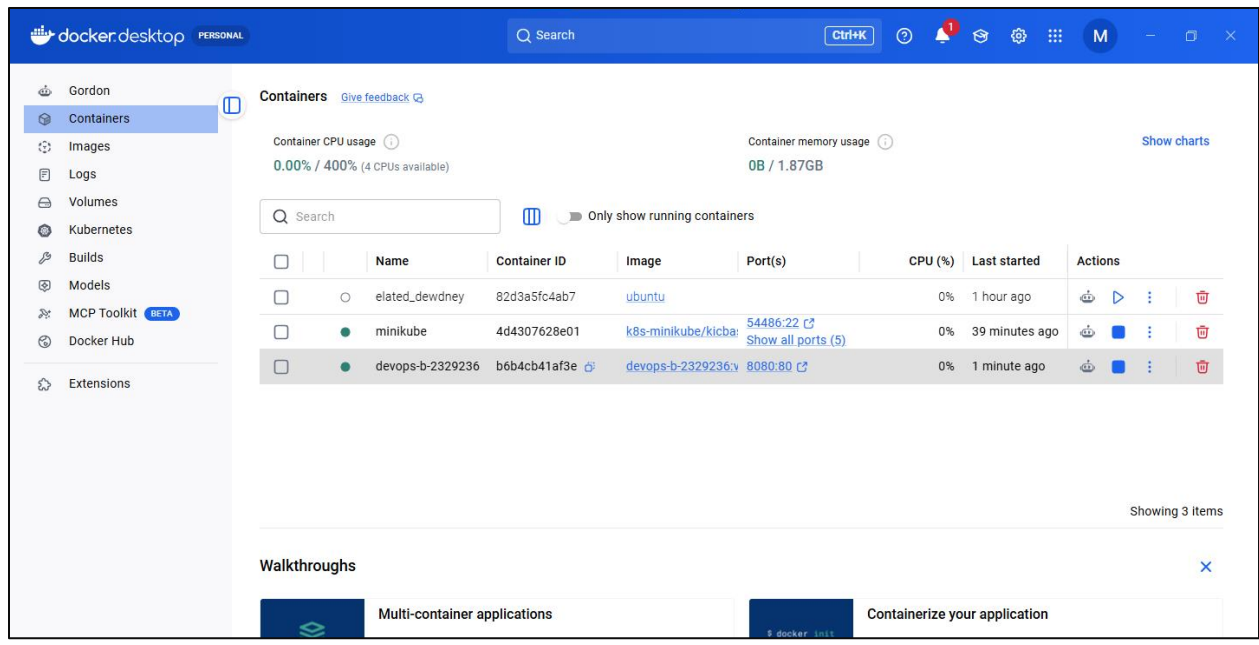
The screenshot shows the Docker Desktop interface. The 'Images' tab is selected, and the 'Local' section is active. The 'Images' list shows three images:

Name	Tag	Image ID	Created	Size	Actions
ubuntu	latest	6e92c4fd8dfd	8 days ago	100.87 MB	📄 ▶ ⋮ 🗑️
gcr.io/k8s-minikube/kicbase	v0.0.51	2c31ef2958e5	1 month ago	1.33 GB	📄 ▶ ⋮ 🗑️
devops-b-2329236	v1	7a6ea773cd95	3 minutes ago	62.94 MB	📄 ▶ ⋮ 🗑️

The 'Walkthroughs' section shows two walkthroughs:

- [How do I run a container?](#) (6 mins)
- [Run Docker Hub images](#) (5 mins)

The 'Upgrade plan' button is visible at the bottom left.



```

PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS
PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> docker build -t devops-b-2329236:v1 .
=> => extracting sha256:e2de96513ba9eb53b431787ec8a65cdde380ac4772a3e4c4b714dcfde2a102b5          0.2s
=> => sha256:e76228b47809e8a4bd37a6ba52a04d282d8c72dd0ff9e8b83fd6318631abd52 1.40kB / 1.40kB 2.4s
=> => extracting sha256:d9aae54b58315e84b9c3ac5f00bad6610e0e28ad9ff28fcd694647e0be13131          0.1s
=> => sha256:e72112c14215406dc7f23ef8e5cfdffdeb8662176167382ad64ec86f52b7a88e / 20.53MB          5.8s
=> => extracting sha256:6c53db2a666ed8f8fad8d9982e30cf24be085036c17b27591665ca25e1ba40c          0.0s
=> => extracting sha256:745dfb2690dd8627e1593941cacc756f14943df7930a7478beb4e808c5695cb9          0.0s
=> => extracting sha256:9a9a644fdd6a3681d43dace44c8cfef374cfd7b24f433113379ea483692498a8          0.0s
=> => extracting sha256:64c8194480fee2c03e01cc561cf5542733627d2c7de493a028139677025f975c          0.0s
=> => extracting sha256:e76228b47809e8a4bd37a6ba52a04d282d8c72dd0ff9e8b83fd6318631abd52          0.0s
=> => extracting sha256:e72112c14215406dc7f23ef8e5cfdffdeb8662176167382ad64ec86f52b7a88e          0.7s
=> [2/2] COPY site/ /usr/share/nginx/html/                                                    0.2s
=> exporting to image                                                                            0.3s
=> => exporting layers                                                                            0.1s
=> => writing image sha256:7a6ea773cd95174837db73799671f212831b0b32e4962b206e4f283c2a7420b4 0.0s
=> => naming to docker.io/library/devops-b-2329236:v1                                          0.0s

1 warning found (use docker --debug to expand):
- ConsistentInstructionCasing: Command 'FROM' should match the case of the command majority (uppercase) (line 1)
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> docker run -d --name devops-b-2329236 -p 8080:80 devops-b-2329236:v1
b6b4cb41af3e52295f5b1a3f61b512318e99d4beb1a2412efa7e344fc644b5dd
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> docker ps
CONTAINER ID    IMAGE                                COMMAND                                            CREATED        STATUS        PORTS
b6b4cb41af3e    devops-b-2329236:v1                "/docker-entrypoint..."                       27 seconds ago Up 26 seconds 0.0.0.0:8080->80/tcp, [
::]:8080->80/tcp
4d4307628e01    gcr.io/k8s-minikube/kicbase:v0.0.51        "/usr/local/bin/entr..."                      37 minutes ago Up 37 minutes 127.0.0.1:54483->2376/tcp, 127.0.0.1:54484->5000/tcp, 127.0.0.1:54485->8443/tcp, 127.0.0.1:54487->32443/tcp
minikube
○ PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236>

```

Q2(c) — Kubernetes NodePort Service

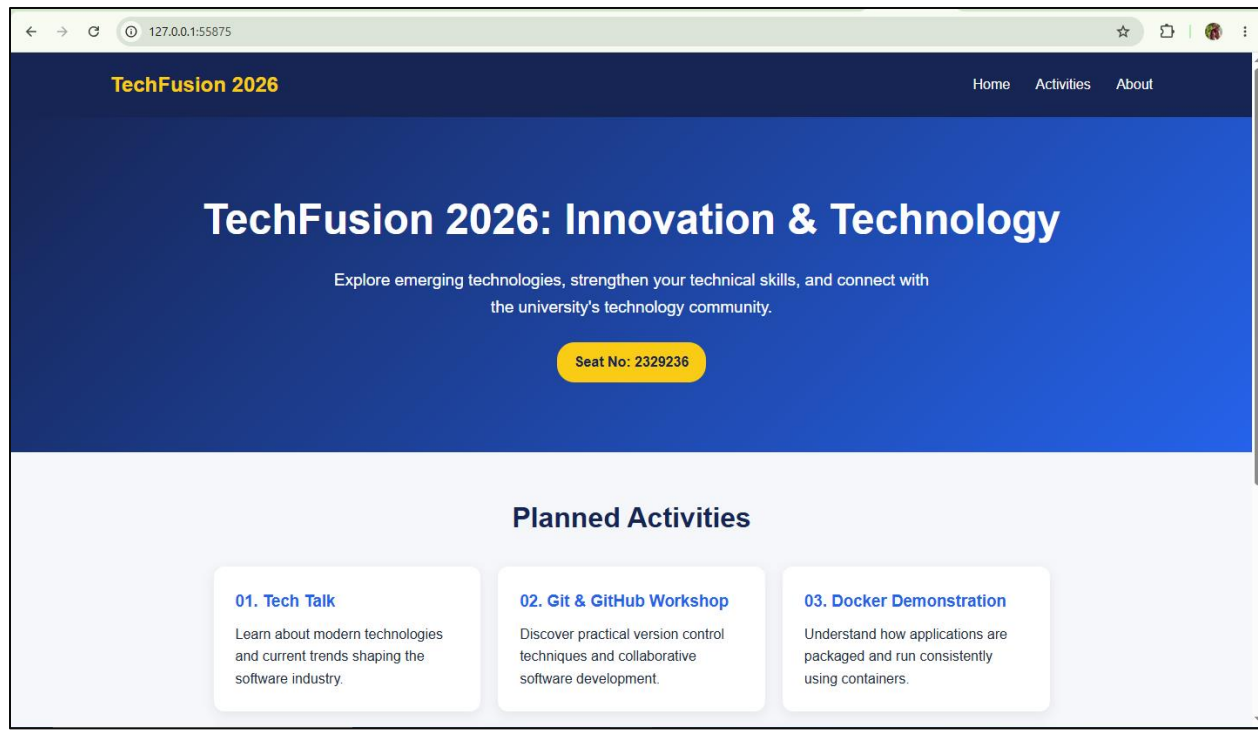
```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\WINDOWS\system32> minikube image load devops-b-2329236:v1
PS C:\WINDOWS\system32> cd C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236
PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236>
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> kubectl apply -f deployment.yaml
deployment.apps/devops-b-2329236 created
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
devops-b-2329236-75b599b87b-7fd4n  1/1     Running   0           98s
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> kubectl apply -f service.yaml
service/devops-b-2329236 created
● PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> kubectl get service
NAME              TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
devops-b-2329236  NodePort    10.107.245.253 <none>        80:32540/TCP     15s
kubernetes        ClusterIP   10.96.0.1      <none>        443/TCP          53m
○ PS C:\Users\muskan\Desktop\DevOps-Finallab\SetB_Muskan_2329236> minikube service devops-b-2329236 --url
http://127.0.0.1:55875
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```



Q2(d) — Kubernetes Scaling

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● PS C:\Users\muskan\Desktop\DevOps-FinalLab\SetB_Muskan_2329236> kubectl scale deployment devops-b-2329236 --replicas=2
deployment.apps/devops-b-2329236 scaled
● PS C:\Users\muskan\Desktop\DevOps-FinalLab\SetB_Muskan_2329236> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
devops-b-2329236-75b599b87b-7fd4n  1/1     Running   0           10m
devops-b-2329236-75b599b87b-pmwfm  1/1     Running   0           10s
● PS C:\Users\muskan\Desktop\DevOps-FinalLab\SetB_Muskan_2329236> kubectl get deployment devops-b-2329236
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
devops-b-2329236    2/2     2             2           10m
● PS C:\Users\muskan\Desktop\DevOps-FinalLab\SetB_Muskan_2329236> kubectl delete pod devops-b-2329236-75b599b87b-7fd4n
pod "devops-b-2329236-75b599b87b-7fd4n" deleted from default namespace
● PS C:\Users\muskan\Desktop\DevOps-FinalLab\SetB_Muskan_2329236> kubectl get deployment devops-b-2329236
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
devops-b-2329236    2/2     2             2           11m
○ PS C:\Users\muskan\Desktop\DevOps-FinalLab\SetB_Muskan_2329236> 
```